PRE-BOOTCAMP READINESS SESSION

AI in Biomed Boot Camp

Getting everyone set up on AWS — GW Research Gateway

60 minutes · Rocky Linux 8 Remote Desktop · Jupyter + Ollama

CONTEXT

Why we're on AWS today

The situation

- Pegasus GPFS storage failed and won't be fixed before our fixed dates.
- Rescheduling isn't an option — our job is to make it work now.
- GW Research Gateway (AWS) gives everyone a full Linux desktop today.
- Pegasus was always a vehicle, not the point — the AI/biomed skills are.

What carries over

- ✓ Jupyter notebooks
- ✓ PyTorch / Torch
- ✓ Ollama + local models
- ✓ bash / CLI workflows
- ✓ Python environments (venv, pip)

Only Slurm / HPC-scheduler specifics don't transfer.

RUN OF SHOW

The hour at a glance

Slow steps (approval, ~15-min launch, model download) run in the background while we teach.

0:00–0:05	**Welcome + launch instances**	Kick off the ~15-min provisioning clock immediately
0:05–0:20	**While it spins up — teach**	Cloud at GW · access model · Linux basics · Google Colab vs Jupyter
0:20–0:35	**Hands-on setup**	Log into the desktop, stand up Jupyter, start the model pull
0:35–0:50	**Notebooks + Ollama**	Run ollama_aws.ipynb · models · when to use notebooks
0:50–1:00	**Wrap-up + readiness check**	Troubleshooting · RTShelp / Office Hours · confirm everyone's green

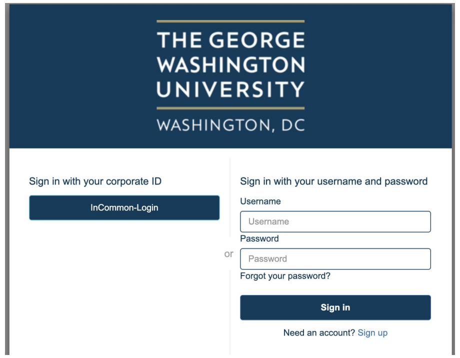
Launch first, talk second

0:00–0:05

Screen-share this click-path — everyone launches together, then we teach while it builds.

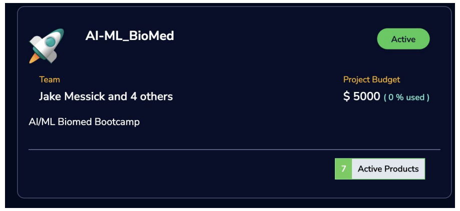
1 Log in

rg.arc.gwu.edu ⟶ InCommon ⟶ Sign in with GW



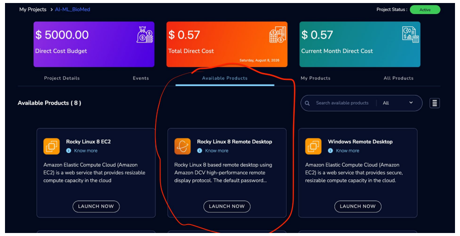
2 Pick the project

Select AI-ML_BioMed



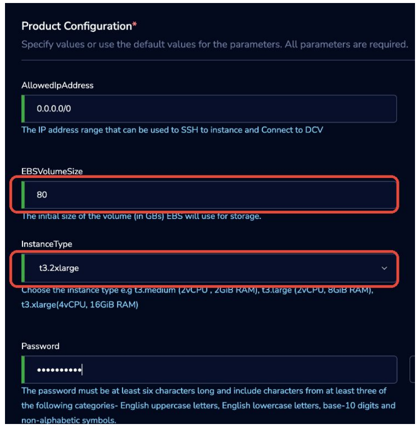
3 Pick the product

Rocky Linux 8 Remote Desktop



4 Configure + launch

80 GB · t3.2xlarge · Launch Now



≈15 minutes to provision. We start every attendee's instance now — no one sits watching a spinner.

THE ONE THING THAT CAN'T WAIT

Admin approval is the hidden blocker

First-ever login shows "requires admin approval." That's normal — but until it's granted, an attendee can't launch anything. Handle it before the clock matters.

1. **Have attendees do their FIRST login the day before**

   So approvals are already granted when the session starts.

2. **Approver on standby at minute 0**

   Someone with admin rights ready to approve any stragglers live.

3. **Confirm everyone is on AI-ML_BioMed beforehand**

   No project membership = nothing to launch.

While it spins up (≈15 min of teaching)

0:05–0:20

1 Where cloud fits at GW

Pegasus (HPC/GPU), Raptor (HTC/HTCondor), Research Cloud (persistent). Why today's skills transfer to all of them.

2 How access works

InCommon sign-in, projects & budgets, and the $5,000 AI-ML_BioMed project you just launched into.

3 Linux & the terminal

What a shell/bash is, tab completion, command history — and why CLI still runs HPC and cloud.

4 Notebooks & Google Colab

What notebooks are good for, and how Jupyter compares to Colab (next slide).

DURING THE SPIN-UP

Google Colab vs. Jupyter on your cloud desktop

Google Colab

- Zero setup — runs in the browser
- Free (limited) GPU, Google-hosted
- Great for quick demos and sharing
- Sessions time out; data lives in Drive
- Limited control over the environment

Jupyter on the cloud desktop

- Full control of packages & environment
- Persistent project storage
- Your own compute (t3.2xlarge)
- Same workflow you'll use in the bootcamp
- Mirrors real HPC / research habits

Bottom line: Colab is perfect for trying an idea; the cloud desktop mirrors what you'll actually run on.

Get into the desktop and stand up Jupyter

0:20–0:35

Log into the desktop

- A new browser window opens with your desktop.
- Log in with the password you created at launch.
- Tip: click in the window and drag upward to reveal the password box.

Then, in a terminal

```
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

Start the model pull early

As soon as Jupyter opens, open **`ollama_aws.ipynb`** and run the model-pull cells right away.

The download runs in the background while we talk through notebooks — so the model is ready before the bootcamp, not during it.

Full path:

```
/home/ec2-user/studies/ProjectStorage/ollama_aws.ipynb
```

Notebooks + Ollama

0:35–0:50

Notebooks

- Run ollama_aws.ipynb end to end.
- What notebooks are good for — and when a plain script is better.
- How notebooks relate to normal shell workflows.
- Same interface you'll use throughout the bootcamp.

Ollama & models

- Calling Ollama from inside a notebook.
- Available models: frontier vs. open-weight.
- Size & quantization trade-offs (speed vs. quality vs. memory).
- The background pull finishes here — confirm the model responds.

When things go wrong — and where to get help

0:50–1:00

Login fails

Re-check InCommon / GW sign-in; first login needs admin approval.

Instance won't launch

Confirm AI-ML_BioMed membership, budget, and 80 GB / t3.2xlarge config.

Jupyter won't start

Activate the venv first (source my_new_env/bin/activate), then re-run.

Package or model missing

Re-run the pip / model-pull cells; check the venv is active.

Where to get help

RTShelp · Office Hours — bring access working, don't wait until the bootcamp.

Watch the budget

t3.2xlarge bills against the $5,000 project budget. STOP your instance when you're done for the day.

READINESS CHECK

You're ready when...

- ✓ Logged into Research Gateway (rg.arc.gwu.edu)
- ✓ Launched Rocky Linux 8 Remote Desktop (80 GB · t3.2xlarge)
- ✓ Jupyter running from your venv
- ✓ Ran ollama_aws.ipynb end to end
- ✓ Model pulled and confirmed responding in a notebook

Come to the bootcamp with all five green.

APPENDIX

Full setup walkthrough

Annotated, redacted screenshots — a self-serve reference to follow before the bootcamp, or if you fall behind the live demo.

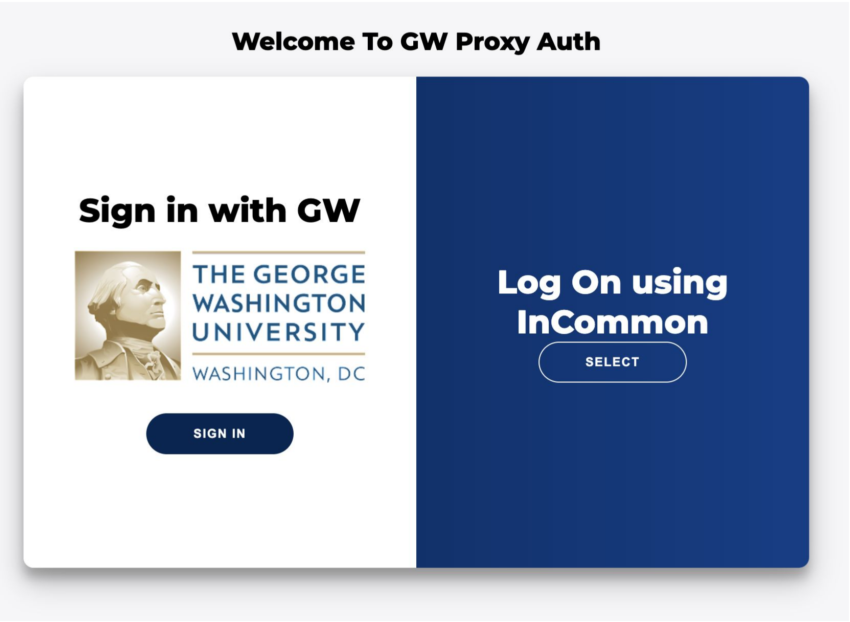
Sign in to Research Gateway

1 Go to rg.arc.gwu.edu

Choose InCommon-Login.

2 Sign in with GW

First-ever login shows “requires admin approval” — that's normal; do this the day before.



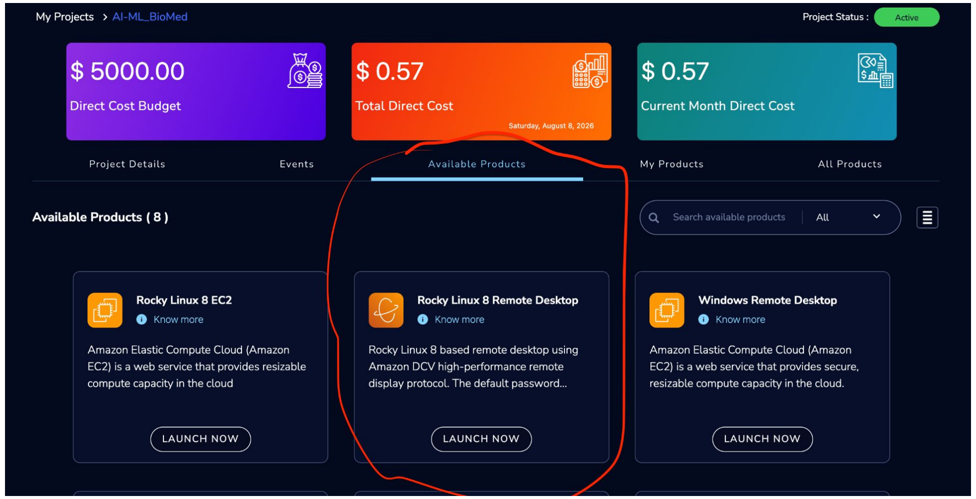
Open the project, pick the product

3 Open AI-ML_BioMed

From My Projects, open the AI-ML_BioMed project card.

4 Available Products → Rocky Linux 8 Remote Desktop

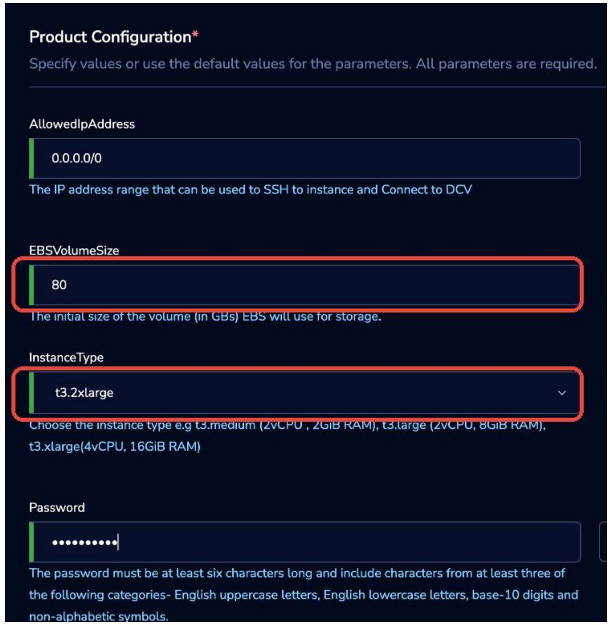
Under Available Products, choose Rocky Linux 8 Remote Desktop (circled).



Configure, launch, and log in

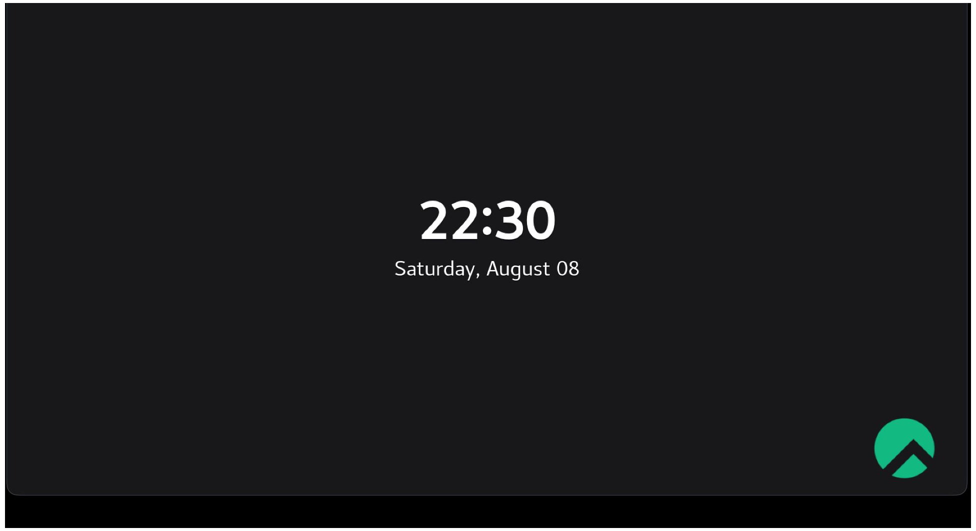
5 Set storage, instance type, password

EBSVolumeSize 80 · InstanceType t3.2xlarge · choose a password ⟶ Launch Now.



6 Log into the desktop

Opens in a new browser tab. Log in with your password (click and drag upward to reveal the box).



Stand up Jupyter and pull the model

Open a terminal on the desktop and run:

```
sudo dnf install python3.12 -y
python3.12 -m venv my_new_env
source my_new_env/bin/activate
pip install notebook
jupyter notebook
```

Then open and run:

```
/home/ec2-user/studies/ProjectStorage/ollama_aws.ipynb
```

Run the model-pull cells first so the download finishes in the background.

A Jupyter notebook opens in a browser on the desktop. The venv must be active before you run jupyter notebook.